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An in silico hierarchal approach for drug candidate mining and validation of natural product inhibitors against pyrimidine biosynthesis enzyme in the antibiotic-resistant *Shigella flexneri*

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ABSTRACT

Shigella flexneri is the main causative agent of the communicable diarrheal disease, shigellosis. It is estimated that about 80–165 million cases and > 1 million deaths occur every year due to this disease. *S. flexneri* causes dysentery mostly in young children, elderly and immunocompromised patients, all over the globe. Recently, due to the emergence of *S. flexneri* antibiotic resistance strains, it is a dire need to predict novel therapeutic drug targets in the bacterium and screen natural products against it, which could eliminate the curse of antibiotic resistance. Therefore, in current study, available antibiotic-resistant genomes ($n = 179$) of *S. flexneri* were downloaded from PATRIC database and a pan-genome and resistome analysis was conducted. Around 5059 genes made up the accessory, 2469 genes made up the core, and 1558 genes made up the unique genome fraction, with 44, 34, and 13 antibiotic-resistant genes in each fraction, respectively. Core genome fraction (27% of the pan-genome), which was common to all strains, was used for subtractive genomics and resulted in 384 non-homologous, and 85 druggable targets. Dihydroorotase was chosen for further analysis and docked with natural product libraries (Ayurvedic and Streptomycin compounds), while the control was orotic acid or vitamin B13 (which is a natural binder of this protein). Dynamics simulation of 50 ns was carried out to validate findings for top-scored inhibitors. The current study proposed dihydroorotase as a significant drug target in *S. flexneri* and 4-tritriacontanone & patupilone compounds as potent drugs against shigellosis. Further experiments are required to ascertain validity of our findings.

1. Introduction

Shigellosis is one of the most communicable gastroenteritis infections caused by the two prominent species of *Shigella* i.e. *S. flexneri* and *S. sonnei*. *S. flexneri* is a gram-negative, enteroinvasive, non-motile facultative anaerobic bacterium and is commonly behind the most shigellosis outbreaks (Chatterjee and Raval, 2019). Shigellosis can be life-threatening, characterized as watery diarrhea and followed by severe dysentery, infecting the human gastrointestinal tract and accompanied by symptoms like fever, and abdominal cramps (Clarkson et al., 2021). It is responsible for ~164 million infections and ~ 1 million deaths annually. ~163 million cases are from developing countries, with 69% of these occurrences in children (<5 age) (Jennison and Verma, 2004). Ranganathan et al. (Ranganathan et al., 2019) recently reported

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that this bacterium causes ~66% of pediatric shigellosis cases in Sub-Saharan Africa and South Asian areas. It spreads via human contact or contaminated food and water ingestion through the fecal-oral route (Chatterjee and Raval, 2019). The currently accepted paradigm indicates that *S. flexneri*, upon reaching the gastrointestinal tract alters the secretion and expression of mucin glycoproteins resulting in the subversion of the mucus membrane (Troeger et al., 2018). The capacity of *S. flexneri* to produce biofilms and antibiotic resistivity renders the management and prevention regimen minimally effective (Olaimat et al., 2017). Currently, many *S. flexneri* outbreaks occur due to the emergence of resistant strains, inflicting an alarming situation for treating shigellosis globally. The rapid increase and widening of the antibiotics resistance spectrum makes *Shigella* infections hard to be adequately controlled by existing prevention and treatment strategies (Kang et al., 2018). Antibiotics like tetracyclines, sulphonamides, fluoroquinolones, co-trimoxazole, azithromycin, pivmecillinam, and ceftriaxone are used against shigellosis (Koo et al., 2017) but the advent of antibiotic resistance in the mutating strains is gradually reducing the efficacy of the existing treatment options (Ranganathan et al., 2019).

Identification of novel drug targets is essential before the drug discovery process. Laboratory screening of macromolecules as drug targets is time-consuming and expensive. On the contrary, development in big data and data mining-based informatics approaches lessens the time and money consumption (Basharat et al., 2021a). The utilization of computational-based methodologies like comparative proteogenomic analysis, pan/core-genomic analysis, structure-based drug designing, etc. reduces the time for target discovery, compared to the conventional laboratory-based experimental practices (Basharat et al., 2021a). In order to combat the menace of resistance, developing new and effective intervention methods/drugs against *S. flexneri* strains is of great necessity. Currently, the use of natural products has become an emerging interest for the prevention and effective treatment of antibiotic-resistant bacterial infections (Barbosa et al., 2020; Rossiter et al., 2017). Therefore, in the current study, we aimed to study the multiple strains of *S. flexneri* and screen therapeutic compounds against an enzymatic drug target. This was done after characterization of the core genome, applying subtractive genomics for drug target selection, and screening of new drug candidates from two natural product (Ayurvedic and Streptomycin) libraries.

2. Material and methods

The current study utilized eight steps for screening drug targets and new natural product inhibitors. These have been listed below:

2.1. Data retrieval

The complete genomes of 179 antibiotic-resistant *S. flexneri* strains were downloaded from the PATRIC database (Davis et al., 2020) on 1 Aug 2021 (URL: https://www.patricbrc.org/view/GenomeList/?and(keyword(Shigella), keyword(flexneri))#view_tab = genomes&filter = eq(antimicrobial_resistance,%22Resistant%22)). The properties of these 179 strains along with their PATRIC accession IDs are summarized in the Supplementary data file. 1. A filter was applied to remove the presence of any redundant sequences. The Universal Protein Resource (UniProt) database was used for the retrieval of human proteome while The Database of Essential genes (DEG database) (Luo et al., 2021) and CEG (Liu et al., 2020) was utilized to investigate the essentiality of drug targets. Latest 2021–3 version of the DrugBank (Wishart et al., 2018) database was utilized to find the druggability potential of shortlisted targets. Furthermore, Streptomycin ($n = 737$ compounds) library (from the ZINC database, URL: https://zinc.docking.org/) and Ayurvedic library ($n = 2103$ compounds) (retrieved from http://ayurveda.pharmaexpert.ru/in) was used for inhibitors screening.

2.2. Pan-genome and resistome analysis

Pan-genomic analysis of these *S. flexneri* strains was performed to calculate the variation found in the genome content (core genome and dispensable genome) using BGPA software, based on parameters described in a previous study (Basharat et al., 2021b). USEARCH clustering algorithm was employed for clustering of only homologous genes present in the pan-genome, with 70% cut-off value using the .fasta files as input. These factors play a dominant role in the growth and pathogenesis of strains. The alignment of these genomes was performed through MUSCLE (Edgar, 2004) tool with default parameters. Dot plot and phylogeny were constructed. UPGMA method was used for tree building. The functional annotation of pan-genome was done through the Cluster of Orthologous Group (COG) (Tatusov et al., 2000) to classify the homologous set of genes.

The concept of pan-genome includes the whole gene content of strains and the idea is now extended to structural variation of these genes, shaped due to certain genomic rearrangements (mobile genes, recombination events, etc). The resistance genes or resistome of *S. flexneri* was then mapped for the above-mentioned genome fractions (pan, core, and unique), using the Comprehensive Antibiotic Resistance Database (CARD) tool (Alcock et al., 2020). Automated BLAST alignment was performed for sequences, against the CARD database, with utilized genes having identity and query coverages greater than 70%. Selection criteria was chosen as perfect and strict hits only.

2.3. Essentiality analysis

Obtained core genes were further used for functional annotation and downstream analysis. The genes playing a significant role in key metabolic pathways of the pathogen are classified as essential genes. To halt the bacterial proliferation and eliminate the pathogen from the body, the identified drug target must be essential for the pathogen's survival. Thus, the non-homologous genes were translated to protein products and further investigated for essentiality analysis using the Database of Essential Genes (DEG) (Luo et al., 2021). The BLASTp of non-homologous genes was performed, with *E*-value cut-off of 10^{-5}. The resultant genes having significant sequence similarities were further validated through the Cluster of Essential Gene Databases (CEG) (Liu et al., 2020). CEG validated the false positive predicted essential gene based on sequences alignment and functionality of genes. The overlapping genes from both essential databases were shortlisted for drug-target prediction.

2.4. Identification of non-homologous sequences to human and gut proteome

The consequent sequences identified from the above analysis were further used for the identification of novel drug targets against *S. flexneri* species. The drug targets should be non-homologous to human proteome, in order to avoid the cross-reactivity of drug candidates. For that purpose, the obtained sets of translated gene products were subjected to BLASTp with a cutoff value of *E*-value 10^{-3}, gap penalty of 11, and gap extension penalty of 1, against the whole proteome of *Homo sapiens*. Additionally, the retrieved proteins were further analyzed against the normal gut microbial flora to evade the responsiveness of drug targets with human microbial floral proteins (Basharat et al., 2021a).

2.5. Therapeutic target identification

The sequences obtained from the procedure in section 2.4 were further screened for druggability. The BLAST tool was applied against the DrugBank sequence dataset, with an *E*-value of 10^{-5}, to evaluate the significant drug targets for substantial drug development. Proteins having a high frequency of sequence similarities (80% or more), were considered as druggable targets.

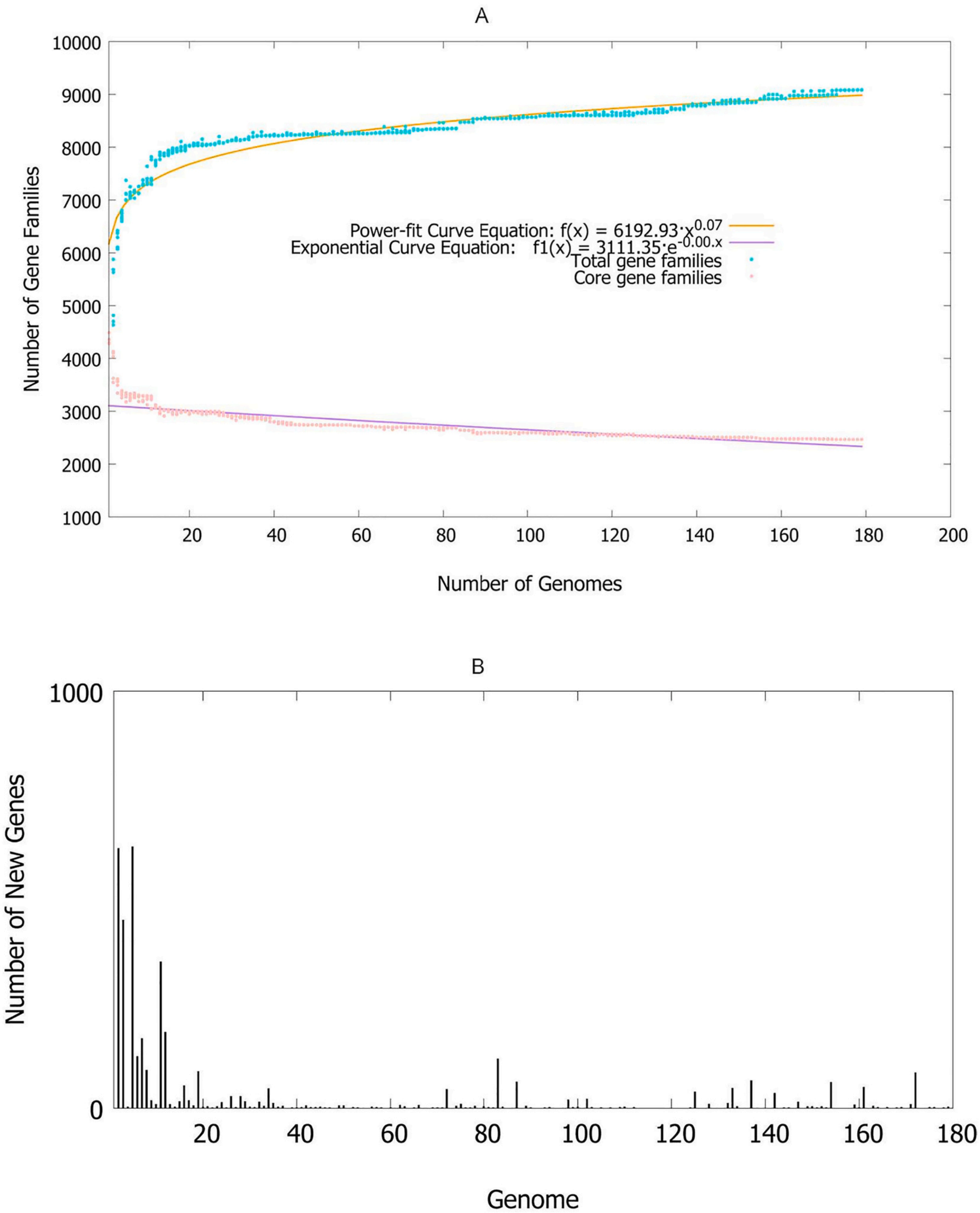

Fig. 1. (A) Pan-genome statistics showing near closing. (B) Number of new genes per addition of new genome.

2.6. Structure modeling and validation

The structure-based drug design requires an understanding of the molecular function and 3D structure of the protein target. The protein shortlisted from the above approach (i.e. dihydroorotase) was searched for homologous structures in the RCSB Protein DataBank (Burley et al., 2019). The structure modeling of protein was done through I-TASSER (Pandit et al., 2006) tool using an iterative approach. Four templates were used, including dihydroorotase (DHO) template belonging to the *Yersinia pestis* (PDB ID: 6CTY), and *Escherichia coli* (PDB ID: 2Z26). Other than these, subtilisin from *Thermococcus kodakarensis* KOD1 (PDB ID: 2Z2Z) and UDP α,α-trehalose-phosphate synthase in complex with glucose synthase from *Burkholderia xenovorans* (PDB ID: 5V0T) was also employed for inferring homology. Decoys were generated for each template and final models were picked using the SPICKER program (Zhang and Skolnick, 2004), after clustering them on the basis of pair-wise structure likeness. Threading and convergence parameters of structure assembly were used to predict the confidence or C-score [range −5 to 2], where a higher value meant better prediction. The final protein model was selected based on its C-scoring and RMSD values. Furthermore, the validation of modeled structure was carried out. Secondary structure evaluation was done using ERRAT, VERIFY3D, and for tertiary evaluation, Ramachandran plot analysis was done through Zlab server (Laskowski et al., 1993).

2.7. Structure-based virtual screening

In molecular docking, the most effective ligand shows the minimal score of docking, against the target protein. Ayurvedic and streptomycin libraries were used for screening drug candidates. Ayurvedic compounds are plant-produced metabolites from the Indian region and have anti-bacterial activity along with health benefits. Broad-spectrum antibiotics produced as metabolites from the bacterial genus '*Streptomyces*' can also be used as prospective drug candidates. The protein structure was used as a receptor while these compounds libraries were used as ligands. The standard docking procedure was applied as: placement = Triangle Matcher, rescoring 1 = London dG, rescoring 2 = affinity dG, and refinement = forcefield, for the molecular docking using MOE (Molecular Operating Environment) software. The topmost docked compounds were selected based on their S-value score, defining the binding of small molecules to protein. Furthermore, the interaction of docked protein-ligand complexes was also analyzed through the MOE program to investigate the hydrogen bonding and hydrophobic interactions between receptor and ligand atoms within a range of 5 Å. MM/PBSA was calculated using parameters: Salt = NaCl, concentration = 0.1, interior dielectic = 4, exterior dielectic = 80, temperature = 300 K.

2.8. Molecular dynamics simulation and ADMET profiling

The most potent compounds identified through molecular docking studies were subjected to molecular dynamics simulation. This was done to evaluate the stability, flexibility, interactions, and inhibitory potential of ligands/drug compounds. Desmond from Schrödinger LLC was used for this purpose. For system preparation, Desmond system builder parameters were set as: OPLS3e forcefield, solvation with TIP3P model, orthorhombic boundary box size (a, b, c distance = 10 Å) calculated through the buffer, and NPT ensemble for 50 ns trajectory at 1.0132 bar pressure and 300 K temperature (Basharat et al., 2021c). Eventually, interaction analysis was executed and the RMSD plot was visualized. RMSF was also calculated using the CABS-Flex server (http://212.87.3.12/CABSflex2/) (Jamroz et al., 2013) with default parameters. In the fluctuation profile using CABS-Flex, Cα atom positions are averaged over the whole trajectory (Jamroz et al., 2014). This helps analyze the unobserved dynamic behavior. RMSF is a gauge of individual residue flexibility, instead of positional changes of the entire protein structure over time. RMSF is, therefore, plotted versus residue number instead of time (as in RMSD).

Importantly, the pharmacodynamics, pharmacokinetics, and toxicity profile of shortlisted top drug-like compounds was determined using the SwissADME tool (Daina et al., 2017). It predicts the absorption, distribution, metabolism, and excretion of compounds. pkSCM tool (http://biosig.unimelb.edu.au/pkcsm/) was used to identify the drug-likeness behavior of compounds, possessing higher penetration and minimum side effects to the human body. Toxicity was predicted through Pre-ADMET (www.bmdrc.org/preadmet) and pkSCM online server. The toxicity profile includes the parameters of toxicity, teratogenicity, neurotoxicity, immunotoxicity, mutagenicity, and carcinogenicity.

3. Results

3.1. Pan and core genome analysis

It is implied that the entire genomic repertoire of an organism cannot be fully understood through few genome sequences. To overcome this, the concept of pan-genome analysis is used in the investigation of the species genomic diversity. It utilizes an entire set of genes from all strains within a genus, species, or clade. These genes are divided into the three sub-entities i.e. core genome (genes found in all strains), accessory genome (genes found in two or more strains), and unique genome (genes present in specific strains only). The concept has previously been applied for drug designing in bacteria (Nishida and Ono, 2020).

The pan-genome of 179 strains (Fig. 1A) consisted of >9000 DNA coding sequences. Among these, 2469 genes were shared by all strains and classified as core genome (only 27% of pan-genome). 5059 genes were identified as accessory genes while 1158 genes were strain-specific (Unique genes; Supplementary data file. 1). The size was calculated using the power-law regression model for pan-genome, where $Y_{pan} = A_{pan}.\ x^{Bpan} + C_{pan}$. Values of A, B, and C were 6192, 0.071, and 3111 respectively. As a result, the pan-genome of *S. flexneri* was identified as almost closed (b value = 0.07). However, due to consistent evolution and horizontal gene transfer, Basharat and Yasmin (Basharat and Yasmin, 2016) have proposed that the pan-genome always remains open for bacterial species.

Moreover, the genome analysis shows the number of least accessory genes found in the strain AUSMDU00008360 (PATRIC accession: 623.1339) i.e. 1527, and a maximum number of accessory genes in strain CTRSIUE-6 (PATRIC accession: 623.305) (2276 genes). No singleton was found in 94 strains while 3 strains had >100 unique genes (the highest number of unique genes was identified as 438 genes in strain CTRSIUE-6). The remaining strains were having variable patterns consisting of 1–10 genes (Fig. 1B). Additionally, 33 strains had missing genes that were present in other strains (Supplementary data file. 1).

3.2. Resistome analysis

Resistance genes are responsible for rendering antibiotics useless or less effective via expulsion of antibiotics from the bacterial cell to bypass the drug action (Rodrigues et al., 2021), altering the antibiotic structure or destroying bonds to render it useless (Abushaheen et al., 2020; Munita and Arias, 2016). These mechanisms pose challenges in treating infectious diseases. Pan-resistome analysis of strains resulted in the identification of antibiotic resistance genes (ARGs) in the core, accessory, and unique genome. Interestingly, in the core genome, 34 genes (mainly mdtA, G, E, H, acrD, emr K, A,) were recognized as resistant to antibiotics such as macrolides, fluoroquinolone, aminoglycoside, tetracycline, cephalosporin, cephamycin, Fosfomycin, aminocoumarin, rifamycin, glycylcycline, monobactam, penem, and phenicol antibiotics. Resistance was conferred through antibiotic target alteration, inactivation, and efflux (Supplementary data file. 2). Three genes i.e. PBP3 (penicillin-binding protein), GlpT (Glycerol-3-phosphate transporter), and EF-TU (elongation factor thermo unstable) were predicted to have SNPs at D350N for first, S357N, E448K for the second, and R234F for the third gene, respectively. These SNPs provide additional resistance power against beta-lactams (cephalosporin, cephamycin, penam), fosfomycine, and elfamycin antibiotics. Additionally, 44 genes from the accessory genome were found to be associated with antibiotic resistance. These involved beta-lactamase genes (CTX-M, TEM, ampC, OXA) and efflux genes (mdt, mdf, AcrS, gadX) providing resistance to carbapenem, aminoglycoside, macrolide, fluoroquinolone, and tetracycline. Double SNPs (Y137H, and G103S) were found in marR gene, indicating association with the tetracycline resistance through the mechanism of resistance-nodulation-cell division (RND), and alteration of antibiotic efflux pump (Supplementary data file. 3). Moreover, the unique genome was identified having 13 AMR genes conferring resistance to fluoroquinolone, glycylcycline, tetracycline, diaminopyrimidine, beta-lactams (cephalosporin, penam), macrolides, and nitrofuran antibiotic mainly involving antibiotic target replacement, antibiotic efflux, and antibiotic efflux mechanisms (Supplementary data file. 4).

AMR profiling is critical for monitoring, risk assessment, and management of infections and this study presents a comprehensive resistome of *S. flexneri.* It is evident that the presence of high scale AMR in *S. flexneri* is associated with poor clinical outcomes (Afroze et al., 2017; Houpt et al., 2021). Poverty and poor sanitation are also associated with shigellosis in poor communities from developing countries (Houpt et al., 2021; Jarzab et al., 2013). Patients need to get early care as the impact of Shigellosis also includes intestinal complications and malnutrition (Kotloff et al., 2018), making it more challenging to treat. Timely treatment with an effective antimicrobial can lessen morbidity, mortality, development of complications, and malnutrition (Khan et al., 2013). Growing resistance to frequently used antimicrobials makes this challenging and heralds the need for the addition of new molecules in AMR treatment drug pipelines.

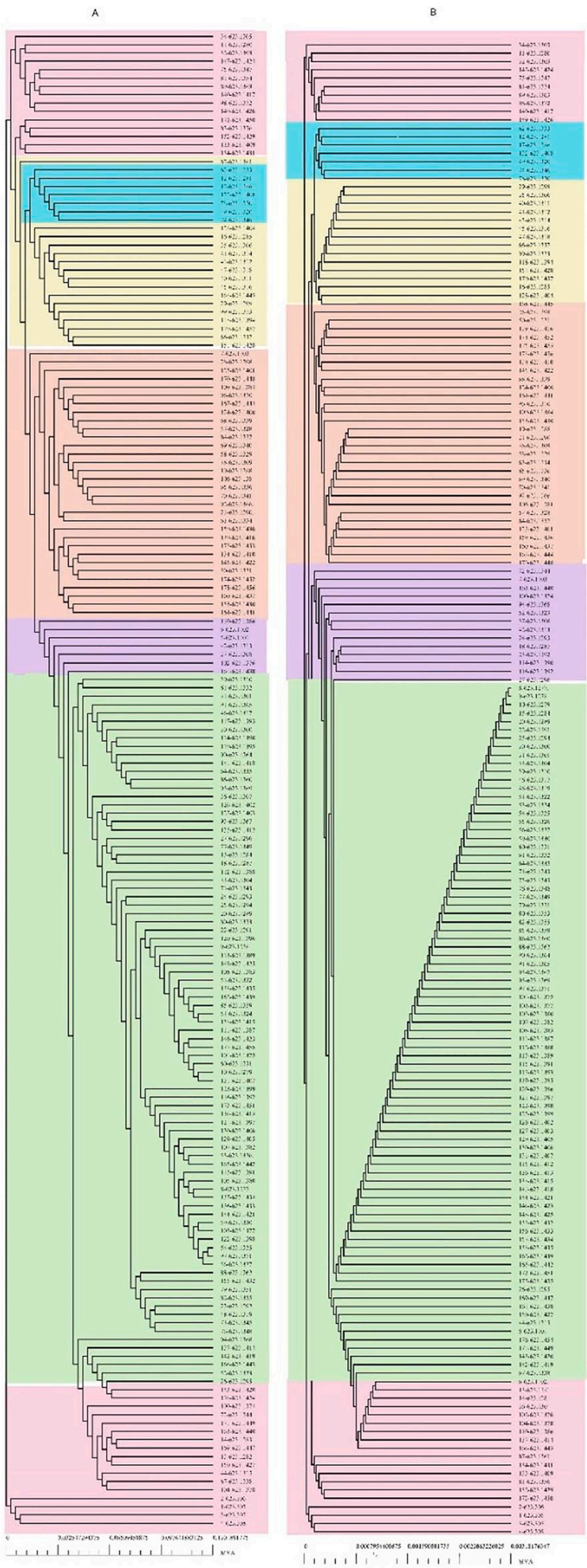

Fig. 2. (A) Pangeome phylogeny (B) Core genome phylogeny. Distance in million years is shown at the bottom.

3.3. Phylogenetic analysis

A phylogenetic tree of 179 strains was generated based on pan and core genome statistics. Distance variation was observed between the sequences of pan and core genome (Fig. 2). Cluster analysis also showed variation in phylogeny, based on both genome fractions. Seven major clusters were color-coded and except for two clusters shown in blue and pink, a major variation in order and number was observed. The distance was very reduced in core genes, compared to pan-genome.

3.4. Functional annotation studies

More accurate pan-genome analysis can be performed by looking into the genes related to specific metabolic pathways. The COG functional annotation for pan-genome analysis of *S. flexneri* revealed that the core genome was enriched in transportation and metabolism pathways of the cell membrane, envelop biogenesis, amino acids metabolism, and translational, ribosomal, & protein biogenesis genes. The accessory genome was mainly found to be enriched in genes responsible for repair mechanism, carbohydrate metabolism, and general functions only. The unique genome was found to have genes involved in transcription, recombination, and repair mechanisms (Fig. 3).

3.5. Identification of essential genes

The core genome, comprising of 2469 CDSs, was used for the identification of novel therapeutic drug targets against *S. flexneri*. The current study utilized two databases for the essentiality prediction i.e. DEG, and CEG. The core genome was first subjected to the DEG database that works on the homology-based search algorithms, it is consists of ~30,000 genes deposited from different ~70 species. It resulted in the identification of 1614 genes as essential for the survival of *S. flexneri*. Secondly, CEG database was used to validate the predicted results from DEG and verify the essentiality of core genes. It applies the coarse evolution method for the functionality prediction of the essential gene from a pre-defined homology-based cluster. This analysis resulted in the prediction of 1433 genes from the 2469 gene set of the core genome. The comparison was drawn between the predicted results and finally, 1418 essential genes found in both databases were shortlisted.

3.6. Non-homologous gene identification

To determine the non-homologous nature of therapeutic targets (both against human and normal gut flora), the selected 1418 essential core genome annotated product was subjected to BLASTp against the whole proteome of humans. From this analysis, only 384 proteins were characterized as non-homologous to human and microbial flora while the rest of the proteins were classified as homologous. These 384 proteins were selected for further downstream analysis.

3.7. Drug target identification

The shortlisted proteins from step 3.6. were subjected to druggability analysis due to their essential, non-homologous nature. The BLAST results against the DrugBank database revealed that only 85 proteins had drug target-like ability. Among these 85 proteins, DHO was selected as a therapeutic target against *S. flexneri*. DHO participates in the third step of pyrimidine nucleotide biosynthesis, where it catalyzes the conversion of carbamoyl aspartate to dihydroorotate (Vitali et al., 2017). This step is a reversible hydrolytic reaction (Peng and Huang, 2014). The function of this protein in building a pyrimidine structure makes it a suitable target for therapeutic purpose. The significant dissimilarity between bacterial and mammalian DHO also enhances its potential as a drug target, as it cannot only disrupt the bacterial growth but can also be used to assess molecular scale AMR response (Lipowska et al., 2019).

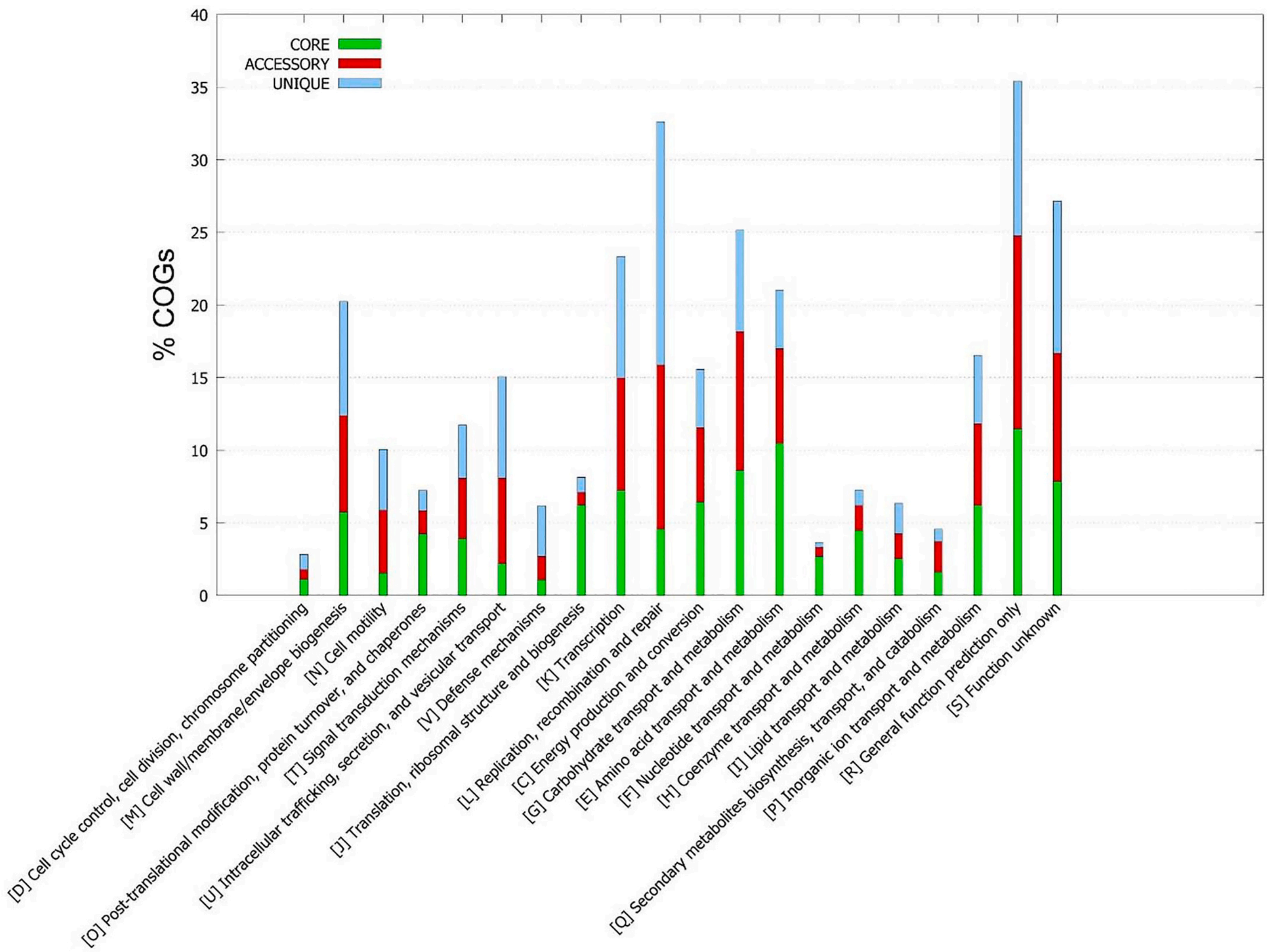

Fig. 3. COG annotated genome fractions of *S. flexneri.*

3.8. Structure prediction and validation

The 3D structure of DHO was modeled using I-TASSER based on the threading approach. Top modeled structure had a TM score = 0.97 ± 0.05, C-score = 1.78, and RMSD of 2.9 ± 2.1 Å (Fig. 4A). It consisted of five sheets, six beta alpha beta units, one beta hairpin, three beta bulges, 15 strands, 18 helices, 18 helix-helix interacs, 21 beta turns and ten gamma turns. The model structure of the protein was passed through ERRAT predicting the quality score as 91.764, VERIFY3D verified the structure of protein showing 100% of residues having a score ≥ 0.2 (Fig. 4B). Moreover, the protein in terms of stereochemical quality showed that 80.8% residues were in the most favorable region, with 17.3% residues in the additionally allowed region and 1.6% in the generously allowed region respectively. Only one residue was found in the disallowed region (Fig. 4C). Ramachandran plot showed 91.531% in the most favored region, 7.1% in preferred regions, and 1.3% in disallowed or questionable regions. These statistics show that the quality of the model was satisfactory for analysis.

3.9. Virtual screening studies

The compounds from Ayurvedic natural product and streptomycin libraries were used for docking-based screening of potential hits. Ayurvedic compounds are plant-produced metabolites from the Indian region and have antibacterial activity along with health benefits (Singh et al., 2021). A dearth of therapies for several diseases, the high cost of new drug manufacturing, and AMR are some reasons for renewed community interest in complementary and alternative medicines. Ayurvedic medicine has a long history (~ 3000 years of practice) based on the understanding of the human body functions as well as the impact of plant substances (Monica et al., 2020). Traditional ayurvedic texts caution against inadequately understood substances. This is the reason that out of ~10,000 plants, only 1200–1500 have been incorporated in the Ayurvedic pharmacopeia and it still needs more extensive research and evidence base (Kumar et al., 2017). For this purpose, we screened 2103 compounds from Ayurveda, against DHO.

In addition, 739 compounds from Streptomyces were also screened. This genus of bacteria, growing in extreme environments, is a reservoir of genes producing a large number of the world's clinical antibiotics (Nepal and Wang, 2019). The core structure can be used as a scaffold for producing new drug molecules as well (Quinn et al., 2020). This heralds a welcome addition to the drying antibiotic pipeline against newly emerging drug-resistant strains.

MOE-Dock was used to determine promising binding poses between the compounds and DHO by aligning and matching the template point triangles with harmonious geometry and chemistry (Indarte et al., 2008), using a centric approach. The DHO atoms stayed fixed during the scanning process and the whole surface was accounted for, during co-ordinate matching. For the binding pattern search, interaction energies between the ligand and the DHO were calculated. For each ligand, 5

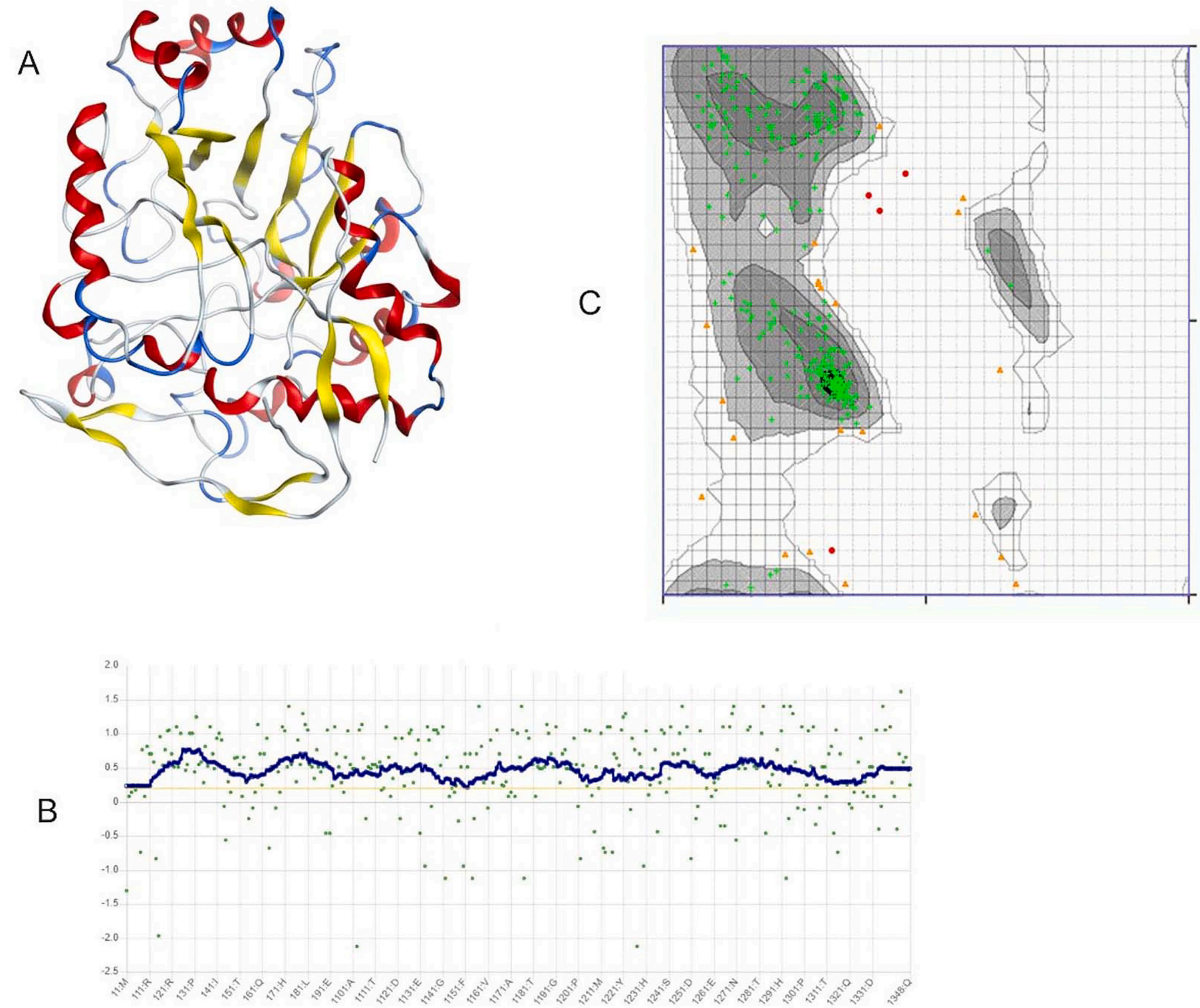

Fig. 4. (A) 3D modeled structure of protein I-TASSER. (B) VERIFY3D profile of protein. (C) Structure validation through Ramachandran Plot showing 91.8% residues in favored region.

poses were generated and scored using London dG function, to determine favorable binding modes. Scoring of these poses was further ranked based on forcefield mediated binding free energy (GBVI/WSA dG) (Labute, 2008). The top model with the lowest energy S-value was taken and interaction was then visualized for finding the details of interacting residues. Based on the obtained S-value score, only one compound from each library was shortlisted as the topmost drug-like hit against DHO protein i.e. 4-tritriacontanone (from Ayurvedic library) and patupilone (from Streptomycin library), having an S-score of −7.76 and − 6.40 respectively. The S-value of the reference compound i.e. vitamin B13/orotic acid was −3.59.

4-tritroacontanone (IUPAC name: tritriacontan-2-one) is a ketone and is found in the seeds of *Achyranthes aspera* Linn., Family Amaranthaceae (Verma, 2016). It has antimicrobial, antioxidant, and anticarminative properties. Its ethanolic extract has shown antibiotic properties against several pathogens like *Bacillus subtilis (*Kaur et al., *2005)*, *Streptococcus mutans* (Yadav et al., 2016), *Salmonella typhi, E. coli,* and *S. flexnerii* (Khuda et al., 2015). Patupilone or Epothilone B (IUPAC name: (1*S*,3*S*,7*S*,10*R*,11*S*,12*S*,16*R*)-7,11-dihydroxy-8,8,10,12,16-pentamethyl-3-[(E)-1-(2-methyl-1,3-thiazol-4-yl)prop-1-en-2-yl]-4,17-dioxabicyclo[14.1.0]heptadecane-5,9-dione) is primarily used as a tubulin stabilizer in cancer treatment (Hussain et al., 2009; Oehler et al., 2012; O'Reilly et al., 2008). It is also categorized as an antibiotic (Drugbank ID: DB03010), which belongs to a class of macrolides. It was initially identified as a metabolite of myxobacterium *Sorangium cellulosum*. Streptome database lists it as a metabolite of *Streptomyces venezuelae* mutant DHS2001 (http://132.230.56.4/streptomedb/compounds_with_organisms/?compound_id = 4934; retrieved on 17 January 2022).

Orotic acid or vitamin B13, a natural inhibitor of dihydroorotase, was used as a reference compound to validate the virtual screening studies and investigate the binding affinity of libraries compounds. Both compounds from the natural product library had lower S-values i.e. better inhibitors of dihydroorotase.

Furthermore, the post-docking analysis for these three compounds (two shortlisted and one reference compound) revealed that Patupilone mediates one hydrogen bond as hydrogen acceptor from Ile120 residue of DHO within the distance of 2.82 Å and has an energy of −1.9 Kcal/mol (Fig. 5A). 4-tritriacontanone was found to be mainly involved in the hydrophobic interaction (Fig. 5B). Vitamin B13 mediated four hydrogen

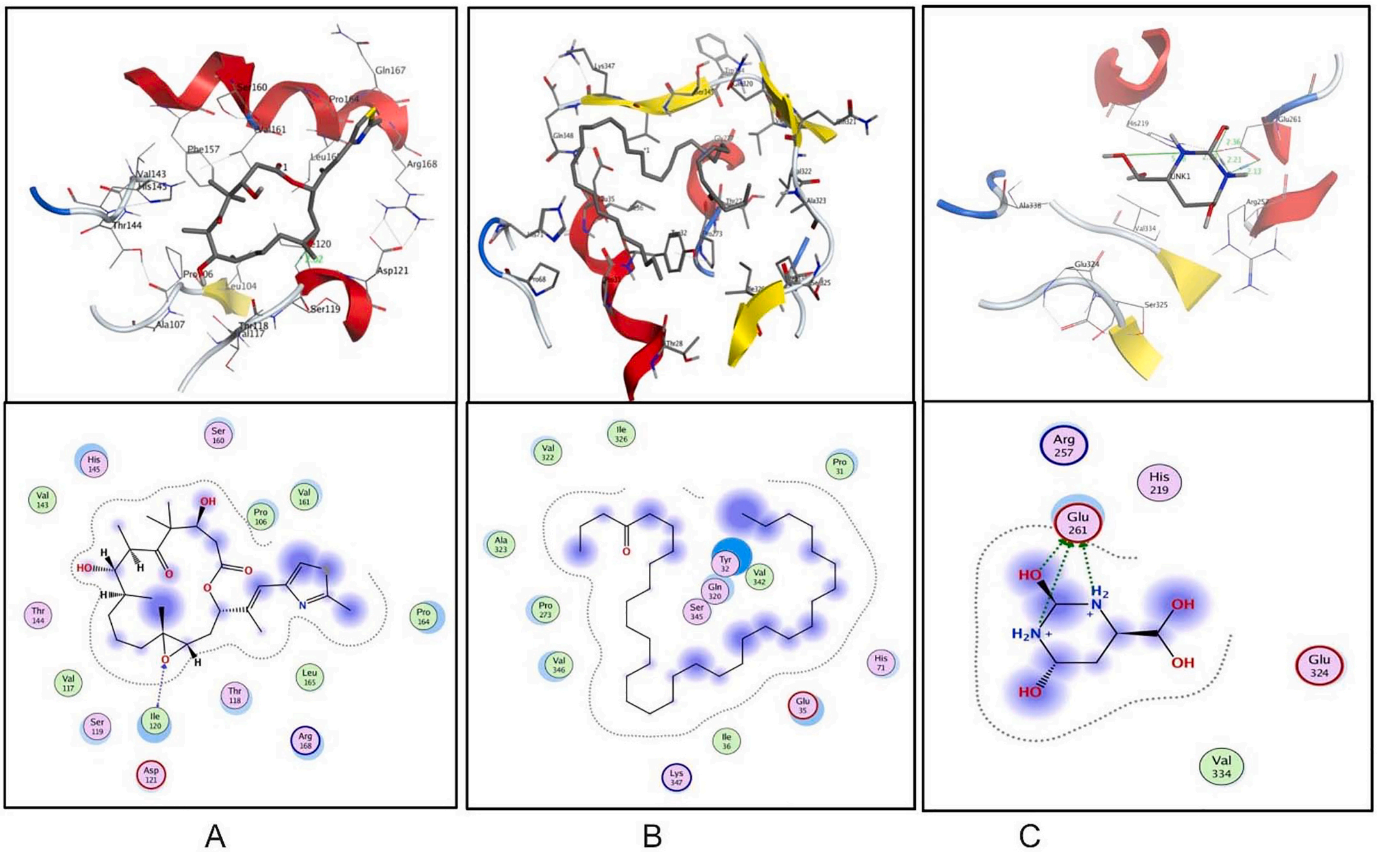

Fig. 5. (A). Docked complex of 4-tritriacontanone (B), interaction between the protein and patupilone while (C) indicates the interaction of orotic acid (vitamin B13) with dihydroorotase.

Table 1
The molecular docking studies of top screened and reference compound.

Compounds	Ligand	Receptor	Interaction	Distance	E (kcal/mol)	S score
4-tritrioctanone	–	–	–	–	–	−3.59
Patupilone	O24	N of Ile120	H-acceptor	2.82	−1.9	−6.40
Vitamin B13	N14	OE1 Glu261	H-donor	3.00	−9.6	−7.76
	N14	OE2 Glu261	H-donor	3.10	−7.9	
	O18	OE2 Glu261	H-donor	3.18	−0.6	
	N20	OE2 Glu261	H-donor	3.01	−10.6	
	N14	OE1 Glu261	Ionic	3.00	−4.5	
	N14	OE2 Glu261	Ionic	3.10	−3.8	
	N20	OE2 Glu261	ionic	3.10	−4.4	

Table 2
MM/PBSA values for the docked complexes.

Compound	MM/PBSA of protein	MM/PBSA of ligand	MM/PBSA of protein-ligand complex
4-tritrioctanone	−21.14	−0.50	−21.04
Patupilone	−21.14	−0.17	−21.04
Vitamin B13	−21.14	0.55	−21.28

bonds and three ionic interactions with Glu216 residue within the range of ~3.0 Å and having binding energies ranging from −0.6 to −10.6 Kcal/mol (Fig. 5C). Table 1 highlights the complete interaction detail of selected compounds with dihydroorotase. Classic MM/PBSA simulated values showed that binding strength was almost equal for both natural products and inferior to that of vitamin B13 (Table 2).

3.10. ADMET profiling of shortlisted drug candidates

It is one of the crucial steps to evaluate the drug safety assessment at the initial step of drug candidate identification to avoid drug toxicity and adverse reaction later on. Vitamin B13 showed a maximum tolerated dose of 1.901 log mg/kg/day, oral rat chronic toxicity of 2.48 log mg/kg/day, LD50 for rat acute toxicity was 1.532 mol/kg. *T. pyriformis* toxicity was 0.285 log μg/L and minnow toxicity of 3.754 log mM. No hepatotoxicity, skin sensitization, or AMES toxicity was observed. High intestinal absorbance of around 84% was seen for patupilone and 86% for 4-tritriacontanone. Both compounds showed significant lead and drug-like properties, following the Lipinski rule, as shown in Table 3.

CYP3A4 binding was observed for both compounds while none

Table 3
AMDET profile of shortlisted compounds from natural product libraries.

Compounds	Solubility	GI Absorption	BBB permeant	Skin Permeability	Carcinogenic	CacO2 permeability	Hepatotoxicity	Skin sensitization	AMES toxicity	Lipinski rule Violation	Lead like Violation
4-tritrioctanone	−5.436	Low	No	−2.735	No	−1.092	No	Yes	No	1	2
Patupilone	−5.754	Low	No	−3.046	Moderate	0.805	Yes	No	No	1	2

bound to renal OCT2. Renal clearance is thus, not observed for both. Total clearance was 0.56 and 2.16 log ml/min/kg for patupilone and 4-tritriacontanone, respectively. This means 4-tritriacontanone would be cleared more quickly from the body. The mechanism remains yet to be elucidated. The maximum tolerated dose for patupilone in humans was −0.321 log mg/kg/day, oral rat chronic toxicity 1.44 log mg/kg/day, LD50 for rat acute toxicity 3.44 mol/kg, *T. pyriformis* toxicity 0.29 log μg/L, and minnow toxicity 0.96 log mM for patupilone. Hepatotoxicity was also observed. The maximum tolerated dose of 4-tritriacontanone in humans was −0.241 log mg/kg/day, rat chronic toxicity 0.63 log mg/kg/day, LD50 for rat acute toxicity 1.921 mol/kg, *T. pyriformis* toxicity 0.28 log μg/L, and minnow toxicity −5.2 log mM. No hepatotoxicity was observed. It was, however, predicted to bind epidermal growth factors and inhibit cell growth.

Both compounds were found to be non-inhibitors of CYP1A2, CYP2C19, CYP2C9, CYP2D6, and non-permeable to blood-brain barrier, with significant permeability to skin and Caco2 human epithelial cell line, used for measuring drug permeability across the intestine. Patupilone was identified as a P-glycoprotein inhibitor while 4-tritriacontanine was not an inhibitor.

3.11. *Molecular dynamics simulation of protein-ligand complex*

A molecular dynamic simulation was then performed for the best-docked models to validate the complex interactions and flexibility. For patupilone, it was observed that the complex was stable at 25 ns with mild fluctuation in ligand while it was observed to be completely stable after 28 ns. His145 and Ile120 showed polar and hydrophobic interaction with ligand for more than 30% of simulation time. For 4-tritriacontanone, only Val346, forming a hydrogen bond, retained contact with ligand for around 30% of simulation time while no residue retained contact for >30% of simulation time for vitamin B13. The average protein-ligand RMSD of 4-tritriacontanone was found below 3 Å. (2.8 Å for protein while 2.4 Å for ligand). The simulation studies for 4-tritriacontanone and protein indicate the stability of complex after 20 ns at 3.6 Å, while for reference vitamin B13, it was found to be highly variant (Fig. 6). However, the RMSF calculated for protein and ligands showed average fluctuation values dwindling around 3 Å (Fig. 7). Reduced flexibility was observed at binding positions. For patupilone, RMSF values were 1 and 2 Å for binding residues Ile120 and His145, respectively. Values were nearly similar for other two ligands as well. For 4-tritriacontanone, value was 0.42 Å for binding residue Val346, with again near similar values for other two ligands.

4. Discussion

Shigella is an inflammatory enteric pathogen producing a global burden of diarrhea. It has been reported that it is responsible for severe community shigellosis, mostly in children under 1 year (Sack et al., 2001). Furthermore, it is responsible for up to 60% of outbreaks of diarrhea (Puzari et al., 2018) due to its ability of easy transmission (via the fecal-oral route and contaminated water) and low infectious dose (>100 bacteria) (Platts-Mills et al., 2015). The molecular diagnostic approaches have identified the substantially greater prevalence of *Shigella* infections which were not observed in past decades. It continues to be the leading cause of high mortality rate. Despite all efforts for shigellosis control, lack of vaccines and emergence of antimicrobial resistance strains are still one of the major challenges for controlling the disease (Ledwaba et al., 2019).

The availability of whole-genome sequences has made it possible to analyze the genome and proteome of the pathogen through computational approaches, within less time, reduced cost, and labor. The emergence of multiple drug resistance and lack of effective drug molecules against *Shigella* have necessitated a search for alternative strategies to further discover potential drug targets against their infections. The advancement in bioinformatics made it possible to identify and develop

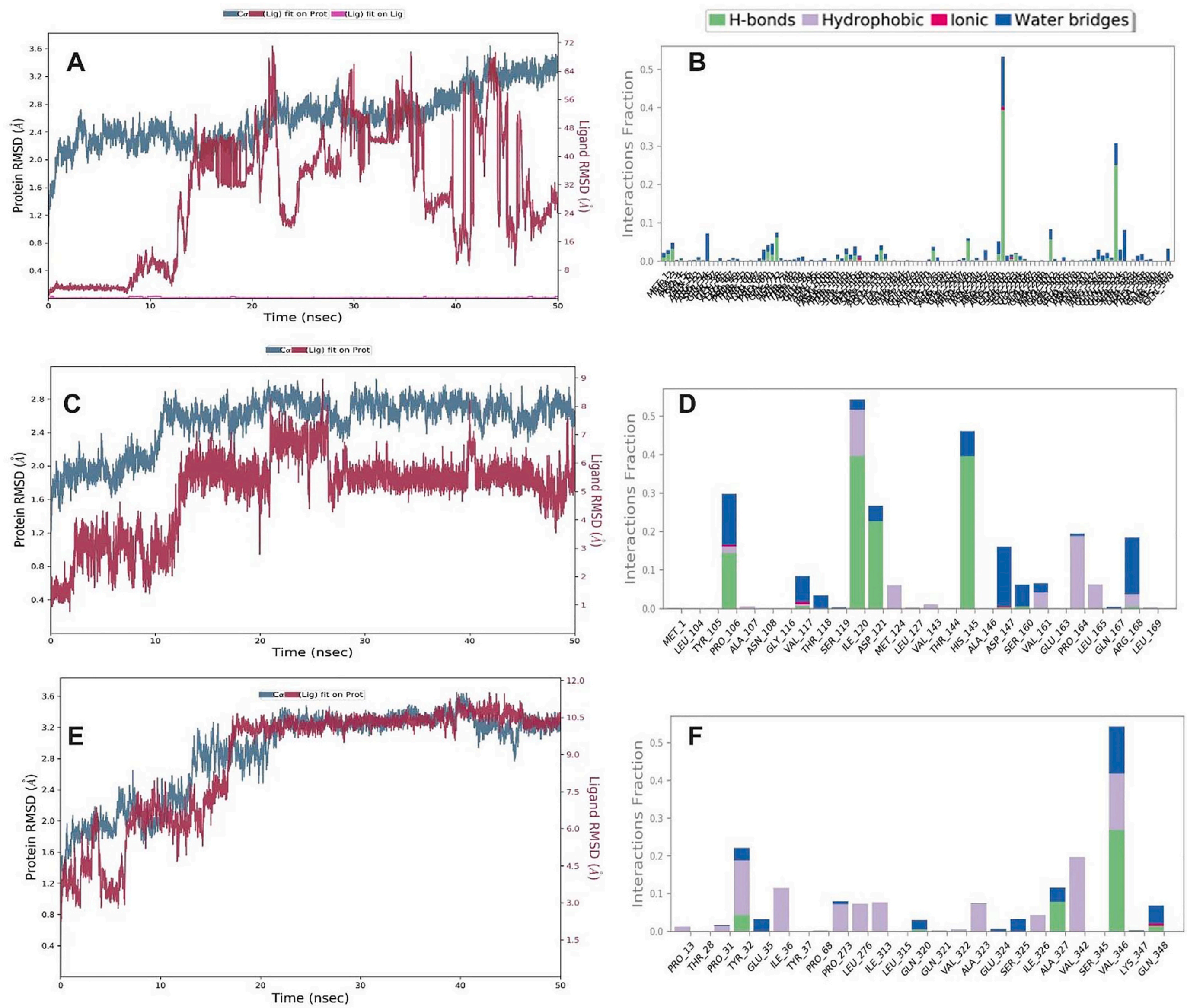

Fig. 6. Molecular dynamic simulation analysis plot showing (A) Vitamin B13 and DHO RMSD, and (B) Vitamin B13 and DHO protein ligand interactions. (C) Patupilone and DHO RMSD (D) Patupilone and DHO protein ligand interactions. (E) 4-tritriacontanone and DHO RMSD. (F) 4-tritriacontanone and DHO protein ligand interactions.

certain therapeutics targets without the utilization of living cells. Therefore, in the current study, we have analyzed the 179 antibiotic-resistant genome strains of *S. flexneri* to understand the resistivity and identify unique drug targets. Openness of the pan-genome showed that the gene pool of *S. flexneri* had no greater limit and it could constantly present novel genes. Previously, open-pan-genome of the ancestral *E. coli* (Park et al., 2019) and related *S. sonnei* (Basharat et al., 2021d) has also been observed. In this specie, ~2469 genes were identified as core fraction, shared by all strains. This number was lower than the related *S. sonnei* strains (2786 genes) but the fraction was 0.48% of the accessory region, compared to 0.38% in *S. sonnei* (Basharat et al., *2021d*). Park et al. (Park et al., 2019) have indicated that evolutionary time should be accounted for while calculating pan-genome. The phylogenetic analysis of core and pan-genome highlighted the variation of clustering in these fractions of genomes for our analysis. The unique genes depict a high probability of horizontal gene transfer but shared genes may have different evolutionary histories. These include vertical inheritance from the ancestral strains or horizontal transfer. Complementation of pan-genome analysis by phylogenetics becomes more essential for pathogenic strains as strain sampling is not even and due to community interest, they tend to be overrepresented in databases. Population genetics should also be combined with pan-genomics to infer details of gene gain/loss dynamics after speciation (Gordienko et al., 2013; Horesh et al., 2021).

CARD (Alcock et al., 2020) aids in mining an ensemble of genes conferring antimicrobial, through homolog identification. Although resistance is thought to be more linked with mobile genetic elements of the bacterium (D'Costa et al., 2006), a significant reservoir of resistance genes was observed for *S. flexneri* chromosomal genome, with resistance genes existing in every fraction of the pan-genome. To the best of the authors knowledge, this is the first pan-resistome report for *S. flexneri*. It provides baseline AMR information for this genus and denotes an effort to illustrate the public health risk due to this phenomenon in *S. flexneri*. Among the mined genes (Supplementary data file. 2, 3, and 4), mdt, arc, beta-lactamases, and sul genes comprised a higher proportion, highlighting that *S. flexneri* had enhanced capability of resistance to beta-lactams, fluroquinolones, and cephalosporin. Although reports exist on how pathogenic strains attain resistance from environmental strains

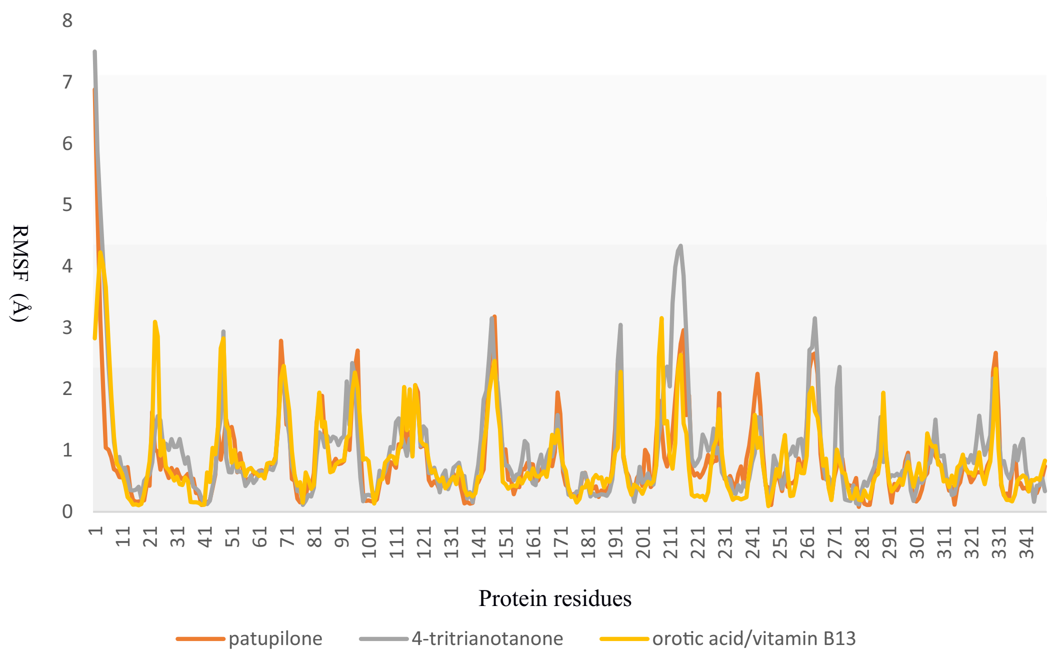

Fig. 7. RMSF plot of top prioritized ligands with DHO.

(Forsberg et al., 2012; Manaia, 2017), additional positive selection pressure by human activities is a severe cause of concern (Dias et al., 2020). This is why it is increasingly becoming important to study new antibiotics and drug molecules from natural sources.

To achieve this aim, we conducted subtractive genomic analysis for *S. flexneri* and 85 proteins were classified as therapeutic drug targets, with DHO involved in the pyrimidine synthesis further selected as a novel therapeutic target for this species. DHO belongs to the cyclic amidohydrolase family and was previously determined as a potential antimalarial and anticancer therapeutic target (Guan et al., 2021a). It has also been reported as an antibacterial target in *Bacillus anthracis (*Rice et al., *2016)*. Natural products like kaempferol, myricetin (Peng and Huang, 2014), and plumbagin (Guan et al., 2021b) have been reported to inhibit it. It was also noted that apart from the active site, compounds also bind at the allosteric site and inhibit DHO activity (Peng and Huang, 2014; Rice et al., 2016). Similar findings were noted for our study, where the active site residues (Thr250, Asp251), predicted by I-TASSER did not make even a single interaction with any of the top binding compounds. Our prioritized compounds made more polar and greasy interactions compared to acidic or basic contacts with DHO. Orotic acid made three hydrogen bonds with the side chain of DHO while patupilone made one hydrogen bond interaction. Ionic interactions were also observed for orotic acid, which were absent in patupilone and 4-tritriacontanone. 4-tritriacontanone and patupilone had better docking scores compared to the reference compound i.e. orotic acid and thus, identified as potent inhibitors of dihydroorotase.

Computational profiling of ADMET in the initial stages of drug discovery is essential to monitor hit-to-lead discovery and afterward, lead-optimization. We pursued to ascertain ADMET knowledge via bioinformatics tools, which were themselves an assembly of several methods. DrugBank listed orotic acid as a non-inhibitor of CYP450 enzyme, non AMES-toxic, and a non-carcinogen. Rat acute toxicity LD50 value was 1.60 mol/kg, whereas software predicted this as 1.5, which was a nearby value. This increases our confidence in the prediction. Both patupilone and 4-tritriacontanone showed lesser rat acute toxicity (LD50 values of 3.4 and 1.9 mol/kg respectively) than orotic acid. Lower toxicity of natural product molecules (resveratrol, xylopic acid, ellagic acid, kaempferol, and quercetin) than FDA-approved drugs like chloroquine has been noted previously (Oso et al., 2021). Similarly, our prioritized compounds also showed less toxicity, and this non-toxic, and drug-like nature needs to be further tested in vitro and in vivo.

Even though molecular dynamic simulation of these protein-ligand complexes highlighted the stability of these protein-compounds complexes after ~20 ns and complemented our docking analysis, along with acceptable ADMET parameters, prediction-based results can still be error-prone as compounds may behave differently in natural settings. Evolution in pathogens and governing of biological systems in humans is influenced by a lot of significant parameters and owing to certain limitations, it is difficult to imitate and simulate the whole settings on a computer system. We can make effort to increase potential parameters and supplement settings with the best possible models. Instead of keeping ligand and protein rigid, we need to ensure flexibility as conformation changes in these moieties are flexible in the solution. This, however, needs lots of computation space and power and a hurdle in obtaining accurate results. Therefore, laboratory validation and in vivo expression of these compounds against *S. flexneri* must be performed as well as investigation of the impact of these compounds on the human immune response, to complement and validate the computational analysis.

5. Conclusion

The increase in the antimicrobial resistance mechanism in pathogens has necessitated the exploration of new and potent drug targets against them. A wide range of genes can be found in the repertoire of species with a lack of understanding of the clustering and resistance mechanisms. This study revealed the significant coherent patterns of resistivity, uniqueness, and similarities in strain genomes of *S. flexneri*. The resistome analysis pointed out the distribution of the AMR genes across the genome fraction, with many genes showing high resistance to cephalosporins, aminoglycosides, and tetracycline. DHO was selected as a drug target and two drug candidates were shortlisted from selected natural product libraries. ADMET profiling suggests them as useful drug candidates. Further in vitro, animal, and pre-clinical studies are suggested for the validation of these candidates and hence, management of shigellosis. Importantly, the swift drug target identification pipeline can be extended to other target pathogens to pave a better way for the control of transmission of persistent infections.

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CRediT authorship contribution statement

Zarrin Basharat: Conceptualization, Formal analysis, Data curation, Validation, Writing – review & editing. **Kanwal Khan:** Formal analysis, Validation, Writing – original draft. **Khurshid Jalal:** Formal analysis, Writing – review & editing. **Diyar Ahmad:** Visualization, Writing – review & editing. **Ajmal Hayat:** Visualization, Writing – review & editing. **Ghallab Alotaibi:** Conceptualization, Methodology, Visualization, Investigation. **Abdulaziz Al Mouslem:** Methodology, Software, Visualization. **Faris F. Aba Alkhayl:** Investigation, Supervision. **Ahmad Almatroudi:** Investigation, Resources, Supervision.

Declaration of Competing Interest

Authors have nothing to declare that may construe as a potential conflict of interest.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.meegid.2022.105233.

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